

# K. SURESHKUMAR

MEP | Mechanical Engineer

Riyadh, Saudi Arabia   k.sureshkumar199920@gmail.com   [LinkedIn](#)   [Portfolio](#)   [Skill Passports](#)   +9660578575624

- M — Mechanical / HVAC
- E — Electrical / BMS
- P — Plumbing / Fire Protection

## PROFESSIONAL SUMMARY

Versatile MEP Engineer with 4+ years of multi-discipline experience spanning Mechanical, Electrical, and Plumbing (MEP) systems across healthcare, commercial, government, and infrastructure projects. Currently leading MEP and BMS building services operations at Dr. Sulaiman Al Habib Medical Group in Riyadh—one of Saudi Arabia's largest private hospital groups—managing HVAC, electrical, plumbing, fire alarm, CCTV, and access control systems across a large hospital facility with 99.5%+ system availability. Skilled across all three MEP disciplines: HVAC system operation and maintenance (chillers, AHUs, FCUs, VAV, VFDs, cooling towers); LV electrical systems (distribution, UPS, generators, lighting control); and plumbing and fire protection (water supply, drainage, pipework, fire suppression). Experienced in MEP design support, AutoCAD drafting, BOQ preparation, MEP quantity surveying, site supervision, and BMS-integrated controls (BACnet, Modbus, KNX). Holds a B.Tech in Mechanical Engineering and an HND in Building Services Engineering Technology. Seeking MEP Engineer roles across Saudi Arabia, UAE, Qatar, and other countries.

## CAREER HIGHLIGHTS

- ✓ Full MEP operations across healthcare & commercial facilities in Riyadh

✓ LV electrical: distribution boards, UPS, generators & lighting control

✓ BMS-integrated MEP controls: BACnet, Modbus, KNX & SCADA

✓ Certified MEP Quantity Surveyor & BACnet/KNX certified
- ✓ HVAC expertise: chillers, AHUs, FCUs, VAV, VFDs & cooling towers

✓ Plumbing & fire protection: pipework, drainage & suppression systems

✓ MEP design support, AutoCAD drawings & BOQ preparation

✓ B.Tech Mechanical Engineering + HND Building Services Engineering

## PROFESSIONAL EXPERIENCE

### BMS Operation & technical support

#### Dr. Sulaiman Al Habib Medical Group

Sahafah Hospital, Riyadh — Large-Scale Healthcare Facility | 24/7 MEP Operations

Riyadh, Saudi Arabia  
June 2024 – Present

#### Mechanical / HVAC

- Oversee operation and maintenance of all HVAC systems — chillers, cooling towers, AHUs, FCUs, VAV systems, VFDs, and chilled water distribution — ensuring 99.5%+ availability in a 24/7 hospital environment in Riyadh's extremely hot climate.
- Monitor HVAC performance via BMS/SCADA platforms; analyse system trends to identify early-stage faults in chiller performance, AHU fan conditions, and control valve response before failures occur.
- Implement HVAC energy-saving measures including chiller sequencing optimization, VFD speed adjustments, and BMS setpoint fine-tuning to reduce energy consumption and improve plant efficiency.
- Supervise HVAC preventive and corrective maintenance activities; coordinate OEM service visits for chiller and AHU major services.

#### Electrical

- Manage LV electrical systems across the hospital including distribution boards (MDB, SMDB, DB), UPS systems, emergency generator sets, and lighting control systems.
- Oversee CCTV, access control, and fire alarm systems through the IBMS platform — monitoring system status, managing alarms, and coordinating specialist vendors for corrective works.
- Ensure all electrical maintenance activities comply with HSE standards and Permit-to-Work (PTW) procedures.

#### Plumbing & Fire Protection

- Oversee hospital plumbing systems including cold and hot water supply, domestic water pumps, drainage systems, and sanitary installations.
- Coordinate inspection and maintenance of fire alarm and fire suppression systems; ensure all fire protection systems remain operational and compliant with safety regulations at all times.

**Assistant Electrical Engineer**  
**New Solar**

Colombo, Sri Lanka  
Sep 2023 – Nov 2023

- Participated in solar design, installation, and commissioning of solar PV systems for residential and commercial projects — covering electrical wiring, inverter configuration, and system monitoring setup.
- Conducted energy assessments and PV system sizing calculations; prepared BOQs and technical reports for solar project delivery.

**MEP Engineering Intern**  
**Department of Buildings, Sri Lanka**

Batticaloa, Sri Lanka  
Feb 2023 – Jul 2023

- **Mechanical:** Assisted in HVAC system design, ductwork layout preparation, and mechanical equipment schedules for government building projects.
- **Electrical:** Assisted in electrical wiring layout design, single-line diagram preparation, and panel schedules; supported site inspections of electrical installations.
- **Plumbing:** Supported plumbing system design, pipe sizing, and drainage layout preparation; gained hands-on experience in fire protection and drainage installations on site.
- Prepared MEP drawings using AutoCAD, estimated material quantities, and compiled BOQs and technical inspection reports.

**Site Supervisor – Pipeline & Civil Works**  
**BPPE**

Jaffna, Sri Lanka  
Mar 2021 – Oct 2021

- Supervised water pipeline installation works including pipe laying, jointing, pressure testing, and valve installation to MEP design specifications.
- Managed daily manpower, materials, and equipment; enforced HSE compliance and prepared daily inspection reports and progress updates for engineering management.

**MAJOR PROJECTS**

**Final Year Project - Lincoln University College**

- Optimization of HVAC Energy Consumption in a Commercial Building using Smart Control Strategies

**Solar PV Integration — New Solar, Colombo**

- Solar design, installation, commissioning, BOQ, and energy assessment for residential and commercial solar PV projects.

**Government Building MEP Projects — Department of Buildings, Sri Lanka**

- Full MEP design support (M, E, P disciplines), AutoCAD drawings, quantity surveying, and site inspection across multiple government infrastructure buildings.

**EDUCATION**

**Lincoln University College**

Dec 2025 - Present

Bachelor of Technology (B.Tech) – Mechanical Engineering

**University College of Jaffna**

Mar 2019 – Jan 2024

[Higher National Diploma \(HND\) – Building Services Engineering Technology](#)

**Eclipse Education**

Jun 2024 – Sep 2024

Certificate – MEP Quantity Surveying

**Eclipse Education**

Aug 2025

Skills Upgrade – MEP Engineering

**CERTIFICATIONS & TRAINING**

|                                                |                                                 |                                    |                                            |                                                   |
|------------------------------------------------|-------------------------------------------------|------------------------------------|--------------------------------------------|---------------------------------------------------|
| <a href="#">BACnet Basics</a>                  | <a href="#">BACnet Cybersecurity</a>            | <a href="#">ETS6 eCampus – KNX</a> | <a href="#">Project Management Webinar</a> | <a href="#">Safety &amp; Health for Engineers</a> |
| <a href="#">Mechine Learning for Engineers</a> | <a href="#">Facility Manager's guide to BAS</a> |                                    |                                            |                                                   |

**LANGUAGES**

|         |                                  |
|---------|----------------------------------|
| Tamil   | Native                           |
| English | Professional Working Proficiency |
| Hindi   | Conversational                   |
| Arabic  | Basic                            |
| Sinhala | Conversational                   |

**PROFESSIONAL MEMBERSHIPS**

- Engineering Council, Sri Lanka – ECSL Member
- ASHRAE – Student / Associate Member
- KNX Association – Community Member
- Skill Passport – TVEC\_ Sri Lanka